

Automated Dynamic Analysis Signature Generation

Using Large Language Models for Malware Behavior Analysis

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Project Motivation

- ▶ Manual malware signature generation is time-consuming and error-prone
- ▶ LLMs excel at pattern recognition and semantic understanding
- ▶ Goal: Automate CAPEv2 signature generation from sandbox behavioral logs
- ▶ Innovation: First LLM integration with CAPEv2 Python signature generation

Research Questions

1. What is the quality of LLM-generated signatures (precision/recall/F1)?
2. How well do signatures generalize to unseen malware families?
3. How accurate is automated MITRE ATT&CK technique mapping?

Six-Stage Pipeline

1. **Data Acquisition:** Load CAPEv2 sandbox reports
2. **Feature Extraction:** Extract behavioral features
3. **Behavior IR:** Build unified representation
4. **LLM Analysis:** Semantic interpretation
5. **Signature Generation:** Create Python signatures
6. **Evaluation:** Comprehensive assessment

Dataset: 100 malware samples from 10 families (Emotet, Zeus, Trickbot, etc.)

Feature Extraction

- ▶ Process operations (creation, injection, termination)
- ▶ Registry modifications (run keys, startup entries)
- ▶ File I/O operations (creation, execution, overwrites)
- ▶ Network activity (DNS, TCP, HTTP/HTTPS, C2)
- ▶ API call sequences and patterns

Output: 50+ behavioral features per sample

Behavior Intermediate Representation

- ▶ Normalizes CAPEv2 format into unified JSON structure
- ▶ Groups behaviors by category
- ▶ Enables LLM semantic analysis without raw complexity
- ▶ Includes temporal sequencing information

LLM-Based Semantic Analysis

1. Convert behaviors to natural language descriptions
2. Per-sample LLM analysis with threat level assessment
3. Family-level pattern aggregation
4. Cache all results for reproducibility

Benefits:

- ▶ Semantic understanding of complex behaviors
- ▶ Automated threat characterization
- ▶ Reproducible analysis pipeline

Signature Generation & ATT&CK Mapping

1. Generate CAPEv2 Python signatures from LLM insights
2. Map behaviors to MITRE ATT&CK techniques
3. Validate syntax and test coverage

Output:

- ▶ 20 CAPEv2 Python signature files
- ▶ 35+ MITRE techniques identified
- ▶ Signature-to-behavior mappings

Signature Generation Results

- ▶ Total signatures generated: 20 (2 per family)
- ▶ Signature generation success rate: 100%
- ▶ Python syntax validation: 100% pass
- ▶ Families covered: 10/10

Distribution:

- ▶ 5 samples per signature
- ▶ All CAPEv2-compliant format
- ▶ Production-ready for deployment

MITRE ATT&CK Mapping

- ▶ Unique techniques identified: 35+
- ▶ Top techniques:
 - ▶ Process Injection (45 occurrences)
 - ▶ Defense Evasion (38 occurrences)
 - ▶ Persistence (35 occurrences)
 - ▶ Command & Control (28 occurrences)

Insights: All 10 families exhibit multiple ATT&CK techniques (avg. 8.5 per family)

Behavioral Pattern Coverage

- ▶ File Dropping: 81% coverage (412 events)
- ▶ Persistence (RunKey): 72% coverage (156 events)
- ▶ Process Injection: 67% coverage (284 events)
- ▶ C2 Communication: 54% coverage (203 events)
- ▶ Defense Evasion: 48% coverage (165 events)

Key Finding: File operations most common, critical behaviors average 64% coverage

Evaluation Metrics

- ▶ Signature Precision: 86%
- ▶ Signature Recall: 79%
- ▶ F1-Score: 0.82
- ▶ False Positive Rate: 8%

Performance:

- ▶ Complete pipeline: 19 minutes for 100 samples
- ▶ Per-sample average: 11.2 seconds
- ▶ Scalability: Linear with sample count

Cross-Family Generalization

- ▶ Signatures tested on held-out families
- ▶ Average Precision (new families): 77%
- ▶ Average Recall (new families): 65%

Finding: LLM-generated signatures capture generalizable patterns effective on new families

MITRE ATT&CK Mapping Accuracy

- ▶ Overall accuracy: 84%
- ▶ Process Injection accuracy: 95%
- ▶ Registry Modification accuracy: 92%
- ▶ Network Communication accuracy: 89%

Challenge: Discovery and Lateral Movement show lower accuracy (68-71%) due to subtle behavioral patterns in sandbox environment

Key Findings

RQ1: Signature Quality

- ▶ LLM-generated signatures achieve 86% precision, 79% recall
- ▶ Competitive with manual signatures (92% precision, 71% recall)

RQ2: Cross-Family Generalization

- ▶ Signatures generalize to new families with 77% precision
- ▶ Demonstrates pattern reusability across families

RQ3: ATT&CK Mapping

- ▶ 84% accuracy on technique mapping
- ▶ 95% accuracy on process injection techniques

Major Contributions

1. First LLM-based automated CAPEv2 signature generation system
2. 20 production-ready signatures with 86% precision
3. Automated MITRE ATT&CK mapping with 84% accuracy
4. Comprehensive evaluation framework for malware signatures
5. Demonstrated cross-family generalization capability

- ▶ **SOCs:** Rapidly generate signatures for emerging threats (minutes vs. days)
- ▶ **Threat Intelligence:** Automated malware family characterization
- ▶ **Detection:** Detection systems updated quickly with new signatures
- ▶ **Research:** Foundation for LLM-powered security automation

Future Work

1. Enhanced LLM integration with multi-step reasoning
2. Graph neural networks for behavior sequence analysis
3. Broader deployment (SIEM, real-time sandboxes)
4. Evaluation on larger dataset (1000+ samples)
5. Multi-platform signature generation (Yara, SNORT, Suricata)

Thank You

Project Repository:

<https://github.com/ykram051/Automated-Dynamic-Analysis-Signature-Generation>

Questions?